A Simple yet Effective Framework for Few-Shot Aspect-Based Sentiment Analysis

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Zengzhi Wang, Qiming Xie, and Rui Xia*

南京理工大学 文本挖掘实验室
Text Mining Lab, Nanjing University of Science & Technology

Background & Motivation:

- The pre-training and fine-tuning paradigm has become the mainstream framework in Aspect-Based Sentiment Analysis (ABSA).
- Although it has achieved sound performance in the domains containing enough fine-grained aspect-sentiment annotations, it is still challenging to conduct few-shot ABSA in domains where manual annotations are scarce.
- In this work, we argue that two kinds of gaps, i.e., ***domain gap*** and ***objective gap***, hinder the transfer of knowledge from pre-training language models (PLMs) to ABSA tasks.

Our Framework

- **Overview:** *Domain-Adaptive Pre-training* + *Text-Infilling Fine-tuning* + *Fuzzy-matching Prediction Correction*

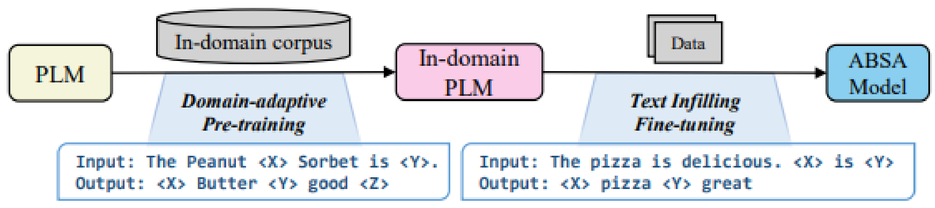

Figure 1: Overview of our FS-ABSA framework. Both *domain-adaptive pre-training* and *text-infilling fine-tuning* are optimized with the text-infilling objective.

- **Domain-Adaptive Pre-training:**
  - ✓ Gururangan et al. introduced domain-adaptive pre-training (DAPT) to continue pre-training RoBERTa on a large-scale domain-specific text for text classification to bridge the domain gap.
  - ✓ In this work, we continue this practice for ABSA, i.e., continuing pre-training the T5 on the in-domain corpus with the text-infilling objective for ABSA, as shown in Figure 1.
- **Text-Infilling Fine-tuning:**
  - ✓ We cast the E2E-ABSA task to a text-infilling task with T5, which we call text-infilling fine-tuning
  - ✓ Here is an example:
    Input: The pizza is delicious. <X> is <Y>
    Output: <X> pizza <Y> great
- **Fuzzy-matching Prediction Correction:**
  - ✓ The predicted aspect terms may differ in tense and case from the ground truths since the target sequence is generated over the whole vocabulary without no constraints.
  - ✓ We propose a heuristic fuzzy-matching-based prediction correction strategies (more details refer to our paper).

Main Results

Table 1: Overall results on 14-Lap and 14-Res under different training data size scenarios (i.e., 32, 128, 512 and Full data). "O" and "D" denotes whether the objective gap and domain gap are filled, respectively. † denotes that our improvements over all baselines are significant ($p \leq 0.05$).

Dataset	Model	O	D	32	128	512	Full
14-Lap	BERT	✗	✗	29.04	48.31	63.10	67.85
	BERT-PT	✗	✓	39.10	55.23	67.96	69.38
	Paraphrase	✗	✗	36.01	57.21	67.59	70.34
	FS-ABSA	✓	✓	**55.62**†	**63.19**†	**70.15**†	**71.16**
14-Res	BERT	✗	✗	46.53	60.00	72.62	76.47
	BERT-PT	✗	✓	56.85	68.04	76.44	79.64
	Paraphrase	✗	✗	52.44	71.28	78.11	81.28
	FS-ABSA	✓	✓	**67.24**†	**74.86**†	**79.43**†	**82.29**

Table 2: Comparison with SOTA under the full data setting.

Model	Backbone	14-Lap	14-Res
SyMux [10]	RoBERTa	70.32	78.68
Paraphrase	T5	70.34	81.28
FS-ABSA (Ours)	T5	**71.16**	**82.29**

Ablation Study

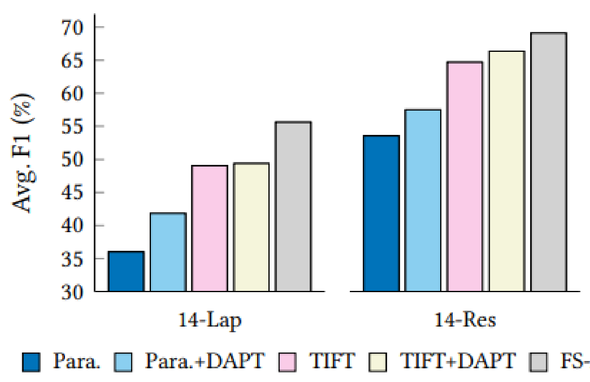

Figure 2: Ablation on *domain-adaptive pre-training* (DAPT), *text-infilling fine-tuning* (TIFT), and *fuzzy-matching-based prediction correction* (FM) involved by our proposed FS-ABSA. All listed models are based on T5.

Quantitative Analysis on Representation Space

Table 3: Cosine similarity of the representations produced by the language model before and after fine-tuning on 32-shot Laptop examples. "O" and "D" denotes whether the objective gap and domain gap are filled, respectively.

Model	O	D	Similarity
TIFT	✓	✗	82.92
Paraphrase + DAPT	✗	✓	84.55
TIFT + DAPT (Ours)	✓	✓	**86.13**

Extension on Low-resource Languages

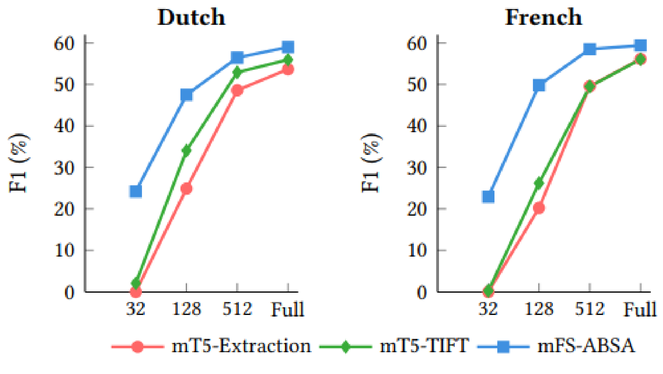

Figure 3: Our results in two low-resource languages under different training set sizes (i.e., 32, 128, 512, and full data).

Feel free to email zzwang@njust.edu.cn.

Our code is available at https://github.com/NUSTM/FS-ABSA

Our pre-trained models are available at https://huggingface.co/NUSTM

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南京理工大学 NANJING UNIVERSITY OF SCIENCE & TECHNOLOGY